Four Volunteers Just Finished a Year of Pretend Life on Mars

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After spending more than a year experiencing what life could be like on Mars, four NASA volunteers have finally stepped outside, marking the end of an ambitious experiment designed to prepare humans for future missions to the Red Planet.

The crew completed their **simulated** Mars journey on October 31, 2026, emerging into the outside world for the first time in more than a year. Although the volunteers never actually left Earth, their extended mission recreated many of the challenges astronauts could encounter while living and working millions of miles from home.

The experiment was part of NASA's Crew Health and Performance **Exploration Analog** (CHAPEA) program, which investigates how people adapt to the demanding conditions of extended space **exploration**. The four volunteers were Matthew Montgomery, James Spicer, Ross Elder, and Ellen Ellis. After finishing their mission, the crew posed together outdoors, celebrating their return from a **simulated** journey to another planet.

Throughout the experiment, the participants carried out tasks similar to those that future Mars explorers may need to perform. These included operating and testing essential systems, caring for crops, conserving limited supplies, and conducting **extravehicular** activities. **Extravehicular** activities, often called spacewalks, are operations astronauts perform outside their spacecraft or habitat. In a Mars simulation, these activities help researchers examine how crews might prepare for and carry out work beyond the safety of their living quarters.

The volunteers also had to manage communication delays, an important challenge for anyone traveling to Mars. Unlike astronauts aboard the International Space Station, who can communicate with Earth almost instantly, future Mars explorers will experience significant delays because radio signals take time to cross the enormous distance between the two planets. These delays mean astronauts cannot always depend on immediate guidance from **mission control**. They must be prepared to solve problems, make decisions, and manage unexpected situations with greater independence.

One of the biggest challenges of sending people to Mars is keeping them healthy, productive, and safe during long periods away from Earth. A crew must be able to manage food, equipment, living conditions, and daily responsibilities while working together in a **confined** environment.

NASA uses **analog** missions to explore these challenges before attempting actual journeys into deep space. An **analog** mission is a realistic simulation that reproduces selected conditions of space **exploration** while allowing scientists to study how people respond in a controlled setting on Earth. The lessons learned from Montgomery, Spicer, Elder, and Ellis will help NASA plan future human **exploration** missions. Their experiences are expected to inform the development of NASA's planned Moon Base and preparations for eventual crewed expeditions to Mars. Although this crew's journey took place entirely on Earth, the

knowledge gained could help address some of the challenges humans will face when they finally travel to the Red Planet.

Word Treasure

analog -- An analog mission is a realistic simulation that copies some conditions of space travel.

extravehicular -- Extravehicular activities are jobs astronauts do outside their spacecraft or home.

simulated -- Simulated means copied or pretended for practice.

confined -- Confined means shut inside a small space.

mission control -- Mission control is the team on Earth that guides astronauts.

exploration -- Exploration means traveling to learn about a new place.

Background you need

NASA is the United States space agency, and one of its goals is to send astronauts to the Moon and, one day, to Mars.

Mars is much farther from Earth than the Moon, so a trip there would take many months and astronauts would have to live in a small habitat with limited supplies.

An analog mission is a realistic practice run on Earth that copies some of the hard parts of space travel, such as delayed messages and living in a confined space.

By studying how volunteers cope, NASA can learn how to keep future crews healthy, safe, and able to work together.

Break it down

LEAD: Four NASA volunteers finished more than a year of simulated life on Mars on October 31, 2026, without ever leaving Earth.

KEY EVENT: The crew completed NASA's CHAPEA experiment, which studies how people adapt to long space missions.

+- WHO: Matthew Montgomery, James Spicer, Ross Elder, and Ellen Ellis
+- WHAT: Operating systems, caring for crops, conserving supplies, and doing spacewalks
+- WHEN: More than a year, ending October 31, 2026

EVIDENCE: The crew handled communication delays and worked in a confined environment, like future Mars explorers will.

OPEN QUESTION: How will the lessons shape NASA's Moon Base and crewed trips to Mars?

Why it matters

Learning how people cope far from home will help keep the first Mars astronauts healthy and safe on a journey that could last years.

Test what you remember

1. When did the CHAPEA crew finish their simulated Mars journey?

(A) November 1, 2026
(B) November 10, 2026
(C) October 31, 2026
(D) October 31, 2025

2. Who were the four volunteers?

(A) Matthew Montgomery, James Spicer, Ellen Ellis, and Kent Hups
(B) Matthew Montgomery, James Spicer, Ross Elder, and Ellen Ellis
(C) Four astronauts from the International Space Station
(D) Matthew Montgomery, James Spicer, Ross Elder, and John Fay

3. What does CHAPEA stand for?

(A) Crew Health and Performance Exploration Analog
(B) Crew Health and Performance Exploration Academy
(C) Center for Human Activity in Extreme Places
(D) Crew Habitat and Planet Exploration Agency

4. What are extravehicular activities?

(A) Messages sent between Mars and Earth
(B) Tests of food grown inside the habitat
(C) Medical checks done inside the habitat
(D) Operations astronauts perform outside their spacecraft or habitat

5. Why will Mars explorers face communication delays?

(A) Radio signals take time to cross the enormous distance between Mars and Earth
(B) The crew refused to answer messages
(C) Mars has no radio towers
(D) NASA turns off radios during missions

6. What will the crew's lessons help NASA plan?

(A) A telescope on Mars
(B) The first hotel on the Moon
(C) A new space station around Venus
(D) Its planned Moon Base and eventual crewed expeditions to Mars

Answer key (try the quiz first!): 1: C 2: B 3: A 4: D 5: A 6: D

Questions to think about

MY THOUGHTS (at least 20 words):